BJÖRN S. SIEPE

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Marburg, Germany
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RESEARCH INTERESTS

- Design of research software and infrastructure
- Metascience of methodological research
- Multivariate time series methods for intensive longitudinal data
- Applied experience sampling research

EDUCATION

Marburg University 10/2022 - present
Doctoral Student

Goethe University Frankfurt 11/2020 - 09/2022
M.Sc. Psychology

Heidelberg University 10/2017 - 09/2020
B.Sc. Psychology

Mannheim University 10/2014 - 08/2017
B.Sc. Economics

Hölderlin-Gymnasium Heidelberg 09/2005 - 06/2013
Abitur (A-levels)

WORK EXPERIENCE

Marburg University 10/2022 - present
Scientific Assistant, Psychological Methods Lab (PI: Daniel Heck)

LOEWE Center DYNAMIC 02/2025 - present
Statistical Consultant

Consulting on conceptualization, study planning, statistical analyses, and statistical programming.

Marburg University 02/2024 - present
Statistical Consultant, Center for Translational Clinical Psychology

Consulting on conceptualization, study planning, statistical analyses, and statistical programming.

Freelance Work 11/2022 - present
Statistical Consultant

Consulting on conceptualization, study planning, statistical analyses, and statistical programming for clinical psychological researchers from multiple research groups and clinics.

Goethe University Frankfurt 11/2021 - 08/2022
Student Assistant, Departments for Psychological Methods

University Hospital Frankfurt 10/2020 - 10/2021
Student Assistant, Horizon 2020-funded Eat2BeNice project (PI: Silke Matura)

Institute for Psychosocial Prevention Heidelberg 10/2019 - 12/2019
Student Assistant

INTERNSHIPS & VISITS

National University of Singapore 02/2026 - 03/2026
Visiting Doctoral Student (PI: Sacha Epskamp)

Research visit in the Department of Psychology, Lab of Methods, Data Science & AI

University of Groningen 09/2024 - 10/2024
Visiting Doctoral Student (PI: Laura Bringmann)

Research visit in the Psychometrics and Statistics Department

Leiden University 03/2022 - 06/2022
Research Intern, ERC-funded WARN-D Project (PI: Eiko Fried)

Working with experience sampling and passive sensor data in a project aiming to build an early-warning system for depression

German Depression Foundation, Research Center 03/2021 - 04/2021
Research Intern, STEADY project (PI: Ulrich Hegerl)

PZN Nordbaden (psychiatric clinic) 02/2020 - 03/2020
Clinical Intern

TEACHING EXPERIENCE

Marburg University 2025

Modelling and data analysis in R (master's level course statistical programming course)

Marburg University 2023 & 2024 & 2025

Critical reading of research papers (bachelor's level course about methodological criteria of research)

Marburg University 2022 & 2023 & 2024

Multivariate Statistics (bachelor's level course about regression and latent variable modeling)

Brainhack School 2023

Introduction to git/GitHub, Introduction to Machine Learning in Python, Introduction to Open Data, Introduction to Data Visualization in Python

Goethe University Frankfurt 2022

Research Methods and Evaluation II (master's level course about latent variable modeling)

PUBLICATIONS

Preprints

2026

8. **Siepe, B. S.**, Bartoš, F., Peikert, A., Heck, D. W., Sarafoglou, A., Boulesteix, A.-L., & Pawel, S. (2026). Why, when, and how to (or not to) preregister a simulation study. *PsyArXiv Preprint.* `https://doi.org/10.31234/osf.io/cxjrb`

7. Huth, K.B.S., Wijnen, S. M. C. E., Skjerdingstad, N., Bartoš, F., Finnemann, A., Ebrahimi, O. V., **Siepe, B. S.**, Steijvers, L. C. J., van Bilsen, C., Bosma, H., Schumacher, L., & Dukers-Muijrers, N. H. (2026). Social connection features predicting loneliness: A longitudinal, interpretable machine-learning analysis. *PsyArXiv Preprint.* `https://doi.org/10.31234/osf.io/mt76k_v1`

2025

6. Haslbeck, J. M. B., Jongerling, J., **Siepe, B. S.**, Epskamp, S., & Waldorp, L. (2025). Model Checking for Vector Autoregressive Models. *PsyArXiv Preprint*
`https://doi.org/10.31234/osf.io/k6uz4`

5. **Siepe, B.S.**, Haslbeck, J.M.B., Kloft, M., Büchner, A., Zhang, Y., Fried, E.I.F., Heck, D.W. (2025). Introducing openESM: A database of openly available experience sampling datasets. *PsyArXiv Preprint.*
`https://osf.io/qfdtb`

4. Bartoš, F., Pawel, S., **Siepe, B. S.** (2025). Living synthetic benchmarks: A neutral and cumulative framework for simulation studies. *arXiv Preprint.*
`https://doi.org/10.48550/arXiv.2510.19489`

3. Pfadt, J. M., Bartoš, F., Godmann, H. R., ..., **Siepe, B. S.**, ..., de Wit, K., Wuth, J., & Wagenmakers, E.-J. (2025). A methodological metamorphosis: The rapid rise of Bayesian inference and open science practices in psychology. *PsyArXiv Preprint.*
`https://doi.org/10.31234/osf.io/ck3js`

2. **Siepe, B.S.**, Kloft, M., Zhang, Y., Petersen, F., Bringmann, L.F., & Heck, D.W. (2025). Using features of dynamic networks to guide treatment selection and outcome prediction: The central role of uncertainty. *PsyArXiv Preprint*
`https://doi.org/10.31234/osf.io/2c8xf`

1. **Siepe, B. S.**, Kloft, M., Aktepe, S. C., & Heck, D. W. (2025). A Robustness Reproduction of Cox et al. (2023). *I4R Discussion Paper Series No. 196.*
`https://hdl.handle.net/10419/308794`

Peer-Reviewed Articles

<u>2025</u>

14. Hehlmann, M. I., **Siepe, B.**, Mink, F., Peña Loray, J. S., Gomez Penedo, J. M., Fritz, J., Moggia, D., Rubel, J. A., & Lutz, W. (2025). Exploring the predictive value of different affect dynamics for psychological treatment outcome. *Psychological Assessment*, 37(11), 656–669
`https://doi.org/10.1037/pas0001384`

13. Kloft, M., **Siepe, B. S.** & Heck, D. W. (2025). The Interval Consensus Model: Aggregating Continuous Bounded Interval Responses. *Psychometrika*, Advance Online Publication.
`https://doi.org/10.1017/psy.2025.10058`

12. Pawel, S., Bartoš, F.[†], **Siepe, B. S.**[†] & Lohmann, A.[†](2024). Handling missingness, failures, and non-convergence in simulation studies: A review of present practices and recommendations. *The American Statistician.*
`https://doi.org/10.1080/00031305.2025.2540002`

11. **Siepe, B. S.**, Tutunji, R., Rieble, C. L., Proppert, R. K. K., & Fried, E. I. (2025). Associations between ecological momentary assessment and passive sensor data in a large student sample. *Journal of Psychopathology and Clinical Science*, 134(8), 912–925.
`https://doi.org/10.1037/abn0001013`

10. **Siepe, B. S.**, & Heck, D. W. (2025). Multiverse analysis for dynamic network models: Investigating the influence of plausible alternative modeling choices. *Methodology.* 21(2), 123–143,
`https://doi.org/10.5964/meth.15665`

9. Cloos, Leonie[‡], **Siepe, B.S.**[‡], Piccirillo, M., Fried, E.I., Wang, S.B., Helmich, M.A., Johnson, S.U., Hoffart, A. & Ebrahimi, O.V. (2025). Accuracy and Consistency of Visual Analog Scales in

[†]Shared second authorship
[‡]Shared first authorship

Ecological Momentary Assessment and Digital Studies. *Collabra: Psychology*, 11 (1): 142735. https://doi.org/10.1525/collabra.142735

8. **Siepe, B. S.**, Rieble, C. L., Tutunji, R., Rimpler, A., März, J., Proppert, R. K. K., & Fried, E. I. (2025). Understanding EMA Data: A Tutorial on Exploring Item Performance in Ecological Momentary Assessment Data. *Advances in Methods and Practices in Psychological Science*, 8(1). https://doi.org/10.1177/25152459241286877

2024

7. **Siepe, B.S.**, Bartoš, F., Morris, T. P., Boulesteix, A., Heck, D. W., & Pawel (2024). Simulation Studies for Methodological Research in Psychology: A Standardized Template for Planning, Preregistration, and Reporting. *Psychological Methods.* Advance Online Publication. https://doi.org/10.1037/met0000695

6. Langener, A., **Siepe, B. S.**, Elsherif, M. M., Niemeijer, K., Andresen, P. K., Akre, S., . . . Jacobson, N. C. (2024). A Template and Tutorial for Preregistering Studies Using Passive Smartphone Measures. *Behavior Research Methods.* Advance Online Publication. https://doi.org/10.3758/s13428-024-02474-5

5. **Siepe, B.S.**, Sander, C., Schultze, M., Kliem, A., Ludwig, S., Hegerl, U., & Reich, H. (2024). Time-Varying Network Models for the Temporal Dynamics of Depressive Symptomatology in Patients With Depressive Disorders: Secondary Analysis of Longitudinal Observational Data. *JMIR Mental Health*, 11:e50136. https://doi.org/10.2196/50136

4. **Siepe, B.S.**, Kloft, M., Heck, D.W. (2024). Bayesian Estimation and Comparison of Idiographic Network Models. *Psychological Methods.* Advance Online Publication. https://doi.org/10.1037/met0000672

3. Westhoff, M., Heshmati, S., **Siepe, B. S.**, Vogelbacher, C., Ciarrochi, J., Hayes, S. C., & Hofmann, S. G. (2024). Psychological flexibility and cognitive-affective processes in young adults' daily lives. *Scientific Reports 14*, 8182. https://doi.org/10.1038/s41598-024-58598-3

2. Rimpler, A., **Siepe, B. S.**, Rieble, C. L., Proppert, R. K., & Fried, E. I. (2024). Introducing FRED: Software for Generating Feedback Reports for Ecological Momentary Assessment Data. *Administration and Policy in Mental Health and Mental Health Services Research*, 1-11. https://doi.org/10.1007/s10488-023-01324-4

2022

1. Talia, A., Georg, A., **Siepe, B.S.**, Gullo, S., Miller-Bottome, M., Volkert, J., ... & Taubner, S. (2022). An exploratory study on how attachment classifications manifest in group psychotherapy. *Research in Psychotherapy: Psychopathology, Process and Outcome, 25*(3), 286-298. https://doi.org/10.4081/ripppo.2022.653

BOOK CHAPTERS

2025

2. Freeborn, L., **Siepe, B. S.** (2025). Unravelling the dynamics of language learning: Insights from temporal network analysis. In Lowie, Rosmawati, & De Wilde (Eds.), *Research Methods in Applied Linguistics* (Vol. 14, pp. 143–166). John Benjamins Publishing Company. https://doi.org/10.1075/rmal.14.08fre

2021

1. Talia, A., **Siepe, B.S.** (2021). Von der Wiege bis in den Behandlungsraum: Zur Manifestation von Bindungsmustern in der Psychotherapie. In Brisch (Ed.), *Bindung und psychische Störungen: Ursachen, Behandlung und Prävention* (pp. 230–247). Klett Cotta.

SOFTWARE & RESEARCH INFRASTRUCTURE

7. **Siepe, B.S. & Haslbeck, J.M.B.** (2026). `VARcheck`: Visual Diagnostic Checks for Vector Autoregressive Models (R Package). Available on GitHub (`https://github.com/bsiepe/VARcheck`). Status: Maintainer, Author
6. **Siepe, B.S.** (2025). `openESM`: Open database of experience sampling datasets Website: openesmdata.org. Status: Lead Developer, Maintainer
5. **Siepe, B.S.** (2025). `openesm`: Accessing the openESM Database (R and Python Package). Available on CRAN (`https://doi.org/10.32614/CRAN.package.openesm`) and PyPi (`https://pypi.org/project/openesm/`). Status: Maintainer, Author
4. **B**artoš, F., Pawel, S., **Siepe, B. S.** & Čala, Petr. `PublicationBiasBenchmark`: Benchmark for Publication Bias Correction Methods (R Package and Website). Available on CRAN (`https://doi.org/10.32614/CRAN.package.PublicationBiasBenchmark`). Status: Author
3. **Kloft, M. & Siepe, B.S.** (2025). `intervalpsych`: Analyzing Interval Data in Psychometrics (R Package). Available on CRAN (`https://doi.org/10.32614/CRAN.package.intervalpsych`). Status: Author
2. Lane, S., Gates, K. M., Fisher, Z., Arizmendi, C., & Molenaar, P. (2025). `gimme`: Group Iterative Multiple Model Estimation (R Package). Available on CRAN (`https://doi.org/10.32614/CRAN.package.gimme`). Status: Contributor
1. **Siepe, B.S. & Kloft, M.** (2024). `tsnet`: Comparing and Visualizing Networks based on Time Series Data (R Package). Available on CRAN (`https://doi.org/10.32614/CRAN.package.tsnet`). Status: Maintainer, Author

INVITED TALKS & KEYNOTES

<u>2026</u>

12. *Methodological research in psychology: Status quo, issues, and potential solutions.* **Keynote**, 1st Interdisciplinary Symposium on Meta Science for Methods Research, University of Zurich (planned for August 2026).
11. *Introducing openESM: A database of openly available experience sampling datasets.* Psychology Lunch Seminar Talk, Nanyang Technological University Singapore (March 2026).
10. *Introducing openESM: A database of openly available experience sampling datasets.* Measurement is the new black consortium meeting (March 2026), online.
9. *Introducing openESM: A database of openly available experience sampling datasets.* Group Meeting of the Oxford Emotional Disorder Genesis and Evolution (EDGE) Lab, University of Oxford (November 2025), online.

<u>2025</u>

8. *Introducing openESM: A database of openly available experience sampling datasets.* Group Meeting of the Oxford Emotional Disorder Genesis and Evolution (EDGE) Lab, University of Oxford (November 2025), online.

7. *Simulation Studies for Methodological Research: State of the Art, Issues, and Potential Solutions.* Colloquium of the Statistics and Psychological Methods Working Unit, University of Münster (June 2025).

6. *Simulation Studies for Methodological Research: State of the Art, Issues, and Potential Solutions.* With Samuel Pawel, Colloquium of the Methods and Evaluation / Quality Assurance Unit, Freie Universität Berlin (April 2025), online.

5. *Why, when, and how (not) to preregister.* Neurophysics Group Meeting, Department of Physics, Marburg University (February 2025).

2024

4. *Exploration of Ecological Momentary Assessment Data.* Measurement is the new black consortium meeting (May 2024), online.

3. *Historical and Recent Trends in the Forecasting Literature.* WARN-D Group Friday Science Meeting, Leiden University (February 2024), online.

2023

2. *Good Practices for Simulation Studies.* ReproducibiliTea Frankfurt (November 2023), online.

2021

1. *Journal of European Psychology Students & Academic Writing Primer.* EFPSA Research Summer School (July 2021), online.

WORKSHOPS

2026

9. *Introducing openESM: A Database of Openly Available Experience Sampling Datasets.* 1h workshop for the Psych #rstats Club, online (April 2026).

2025

8. *Reproducible research in R and Quarto.* 6h workshop for the Graduate School of Education, University of Wuppertal (June 2025).

7. *Research on research methods.* 3h workshop on simulation studies for German students and PhD candidates, with Susanne Frick & Julien Irmer, online (May 2025).

6. *Preregistration in psychology and related fields.* 4h Workshop, with Anabel Büchner, Berlin University Alliance (January 2025).

2024

5. *Introduction to Data Visualization in R.* 1h Interactive Session, Correlaid R Learning Program, Online (November 2024).

4. *Introduction to Dynamic Network Analysis.* 2h Workshop, DYNAMIC Center ECR Day, Goethe University Frankfurt (October 2024).

3. *Analysis of Pre-Post Designs.* 1.5h Workshop, Department of Clinical Psychology and Psychotherapy, Marburg University (March 2024).

2023

2. *Dynamic Network Analysis in R.* 3h Workshop, Department of Clinical Psychology and Psychotherapy, Goethe University Frankfurt (June 2023).

1. *Introduction to Dynamic Network Analysis.* 1.5h Workshop, Department of Clinical Psychology and Psychotherapy, Goethe University Frankfurt (March 2023).

CONFERENCE PRESENTATIONS

2026

16. *Introducing openESM: A database of openly available experience sampling datasets.* Australasian Mathematical Psychology Conference 2026, Singapore.

2025

15. *Simulation studies for methodological research: Status quo, problems, and potential solutions.* Symposium talk, Conference of the DGPs Methods and Evaluation Section 2025, Berlin, Germany.
14. *Combining machine learning with mixed effects models.* Symposium organizer and chair, Conference of the DGPs Methods and Evaluation Section 2025, Berlin, Germany.
13. *Panel Discussion: Quality Standards for Simulation Studies.* Panel discussion organizer and panelist, Conference of the DGPs Methods and Evaluation Section 2025, Berlin, Germany.
12. *Associations between ecological momentary assessment and passive sensor data in a large student sample.* Symposium Presentation, Annual Meeting of the Society for Ambulatory Assessment 2025, Leuven, Belgium.
11. *GIMME, GIMME more (network models): Multiverse analysis for dynamic network models.* Symposium Presentation and Co-Chair of the Symposium, Annual Meeting of the Society for Ambulatory Assessment 2025, Leuven, Belgium.
10. *Simulation Studies for Methodological Research in Psychology.* Symposium Presentation, 7th Joint Statistical Meeting of the German Statistical Working Group (DAGStat), Berlin, Germany.

2024

9. *Bayesian Estimation and Comparison of Idiographic Network Models.* Symposium Presentation, International Meeting of the Psychometric Society (IMPS), Prague, Czech Republic.
8. *Quality Standards for Simulation Studies in Psychometrics.* Panel Discussion Participant, International Meeting of the Psychometric Society (IMPS), Prague, Czech Republic.
7. *Simulation Studies for Methodological Research in Psychology.* Lightning Talk, Virtual Meeting of the Society for the Improvement of Psychological Science (SIPS), online.
6. *Simulation Studies for Methodological Research in Psychology.* Presentation, Virtual Meeting of the Society for Mathematical Psychology, online.

2023

5. *Bayesian Estimation and Comparison of Idiographic Network Models.* Symposium Presentation, Meeting of the German Society for Methods & Evaluation in Psychology (FGME), Konstanz, Germany.
4. *Augmenting Self-Reports with Passive Sensor Data to Understand Changes in Mental Health.* Symposium Presentation, Annual Convention of the American Psychological Society (APS), Washington D.C., US.*
3. *Developing EMA Items to Build an Early Warning System for Depression (WARN-D Study).* Symposium Presentation, Annual Convention of the American Psychological Society (APS), Washington D.C., US.

*Shared first-authorship or coauthor with significant contributions to presentation

2. *Using passive sensor data to understand changes in mental health outcomes.* Symposium Presentation, International Convention of Psychological Science (ICPS), Brussels, Belgium.*

2022

1. *Temporal Dynamics of Depressive Symptomatology.* Presentation, Annual Meeting of the German Society for Psychiatry and Psychotherapy (DGPPN), Berlin, Germany.

POSTER PRESENTATIONS

2025

2. *Taking uncertainty into account is central when using dynamic network features as predictors.* Poster Presentation, Conference of the DGPs Methods and Evaluation Section 2025, Berlin, Germany.

1. *Taking uncertainty into account is central when using dynamic network features as predictors.* Poster Presentation, Annual Meeting of the Society for Ambulatory Assessment 2025, Leuven, Belgium.

SUPERVISION

First Supervisor: 9 Bachelor's, 3 Master's theses
Second Supervisor: 10 Bachelor's, 15 Master's theses

AWARDS

Best Poster Award 05/2025

Awarded at the Society for Ambulatory Assessment Conference 2025 for the poster "Taking uncertainty into account is central when using dynamic network features as predictors"

Student Teaching Award 07/2024

Awarded by the student council psychology at Marburg University

GRANTS

DYNAMIC Center Mini Grant 01/2026

Amount: 4.136,91€ (internal grant). Financing research stay in Singapore

DAAD Research Travel Stipend 08/2024

Amount: 2.259€. Financing research stay in Groningen

DAAD Conference Stipend 07/2024

Amount: 950€. Travel to the IMPS 2024 Annual Meeting

Marburg Research Academy Conference Funding 07/2024

Amount: 1.500€. Co-Applicant on behalf of Open Science Initiative at Marburg University

APS Conference Financial Assistance 03/2023

Amount: 400$

Erasmus Internship Stipend 03/2022

Amount: 1.089€. Financing research stay in Leiden

German Federal Scholarship (Deutschlandstipendium) 10/2021

Amount: 3.600€ (over one year)

PROFESSIONAL TRAINING

Combining R and C++, DAGStat 2025
Linear Algebra for Machine Learning, Imperial College London, Coursera 2022
Modeling the Dynamics of Intensive Longitudinal Data, Utrecht Summer School 2021
Introduction to Statistical Learning, StanfordOnline, edx Course 2021
Practical Time Series Analysis, SUNY, Coursera 2021
Improving your Statistical Inferences, TU Eindhoven, Coursera 2020

ACADEMIC SERVICE

Organizer of Online Lunch Talk Series on Simulation Studies (URL) 2026
Second Chair of the Task Force on Quality Standards for Simulation Studies of the Psychometric Society 2025 - present
Member of Tenure Track Commission, Department of Psychology, Marburg University 2025 - present
Head of the Special Interest Group for Quality Standards for Simulation Studies of the Quantitative Methods Subgroup of the German Psychological Association 2025 - present
Website coordinator, MITNB Consortium (URL) 2024 - present
Member of Doctoral Committee, Department of Psychology, Marburg University 2023 - present
Board Member Open Science Initiative, Marburg University 2022 - present
Co-Organizer Brainhack Marburg 2023 & 2024 & 2025
Co-Organizer Open Science Days Marburg 2023
Co-Host ReproducibiliTea Frankfurt 2022
Board Member Open Science Initiative, Goethe University Frankfurt 2021 - 2022
Student Council Psychology, Goethe University Frankfurt 2021 - 2022

PROFESSIONAL MEMBERSHIPS

Member of the Society for Ambulatory Assessment (SAA) 2025 - present
Member of the German Society for Psychology (DGPS) 2023 - present

COMMUNITY SERVICE

CorrelAid Volunteer 2024 - present
Student tutor at Asylarbeitskreis Heidelberg (refugee aid) 2015 - 2021
Co-host of yearly after-school orientation day, Hölderlin-Gymnasium 2019 - 2021
Involvement with Jugend Rettet (refugee aid) 2017 - 2018
Assistant coach, youth soccer team of ASC Neuenheim 2013 - 2014

MEDIA COVERAGE

Press release by Leiden University on our article "Associations between ecological momentary assessment and passive sensor data in a large student sample" (URL)
Article on the same paper in The Guardian (URL) and Gizmodo (URL), and Interview by The Signal, the newsletter of the Society for Ambulatory Assessment (URL)
Interview on the Center for Open Science Blog about preregistration for simulation studies (URL)

EDITORIAL POSITIONS

Editorial Fellow, Advances in Methods and Practices in Psychological Science 2026 - 2027
Editorial Team, Journal of European Psychology Students 2021 - 2023

REVIEWING

Ad-Hoc reviewer for *Administration and Policy in Mental Health and Mental Health Services Research*; *Behavior Research Methods*; *Brain, Behavior, & Immunity - Health*; *Acta Psychiatrica Scandinavica*; *Advances in Methods and Practices in Psychological Science*; *Child Psychiatry & Human Development*; *Journal of Consulting & Clinical Psychology*; *Psychological Methods*; *npj Digital Medicine*; *Nature Human Behaviour*; *Multivariate Behavioral Research*; *Journal of Open Source Software*; *Cognitive Therapy and Research*; *Journal of Robustness Reports*; *Journal of Mathematical Psychology.*
Reviewer for the Bi-Annual Meeting (2024) of the *German Psychological Society (DGPS)* and the Annual Meetng (2026) of the *Society for Ambulatory Assessment.*